



WORKSHOP

(TURNING EXP-1)

Name- Priyankam Bhuyan

B.TECH CE SEM 2 2024

Roll no-2413007033

January – June, 2025

Experiment 1: To make a job as per drawing

Aim: Knurling, Taper Turning and Face Turning.

Material: Mild Steel

Size: 80 mm x 32 mm

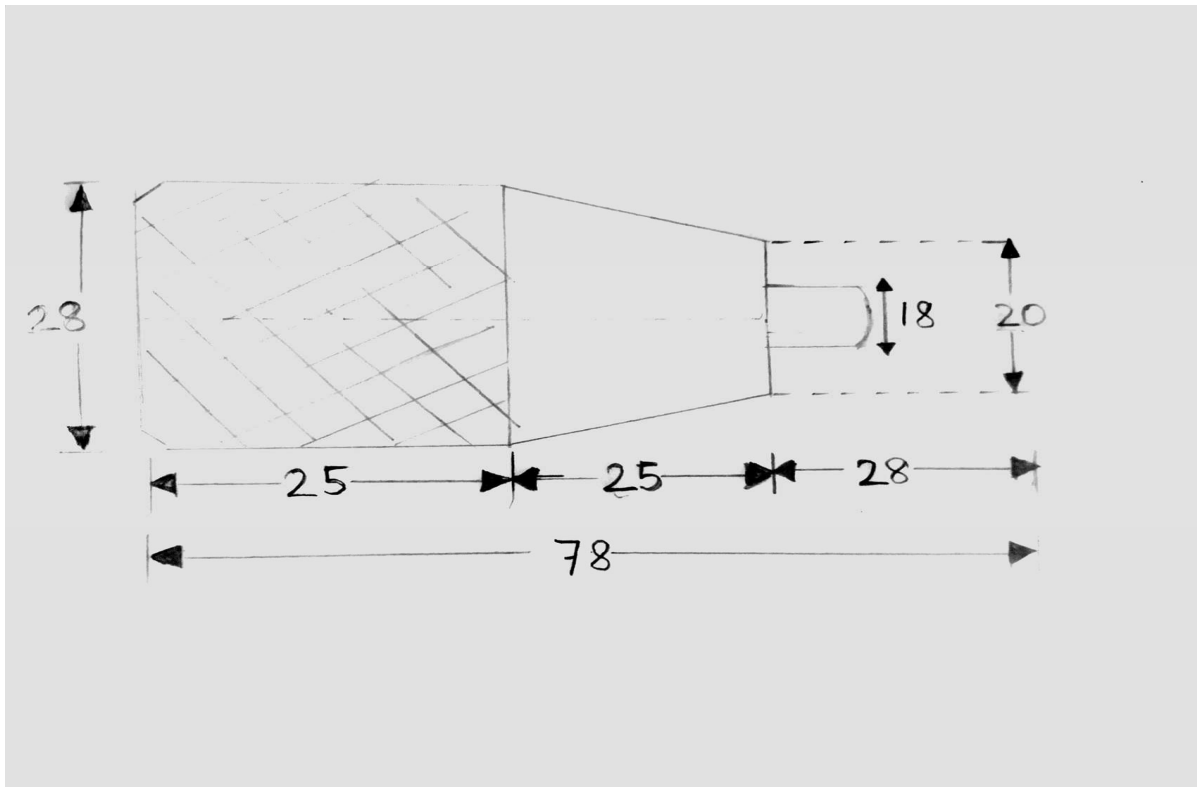
Tolerance: ± 0.06 mm

Tools Required:

- Jaw chuck key
- Tool post key
- Single point cutting tool (High speed steel)
- Cemented Carbide
- Tungsten Carbide
- Double ended spanner
- Vernier Caliper
- Surface Gauge
- Flatline
- Knurling Tool

Job Operation:

1. Job setting
2. Tool setting
3. Facing
4. Plain Turing
5. Knurling
6. Tapering
7. Chamfering
8. Finishing



(All Dimensions are in mm)

Procedure:

1. Firstly, the job is set on jaw chuck.
2. Using the single point cutting tool, the length and radius of the job is reduced to 78mm and 28mm by face turning and plain turning.
3. Now using the vernier caliper 25mm, 25mm, 28mm is marked.
4. In the right side of the job the 20mm and 18mm diameter is marked using the single point cutting tool.
5. Once marked, from the marked 20mm, plain turning is done for 28mm long, upon reaching from 28mm, 25mm is tapered to form a conical type shape.
6. Now from the 18mm that was marked on the job, again we start plain turning till 28mm.
7. From left side of the job, the 25mm area is now knurled using the knurling tool.
8. The pointy edges are now chamfered to rounded edges.
9. The entire job is smoothed out using a sandpaper.
10. Finishing the job.



WORKSHOP

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Experiment 2: To make a job as per drawing

Aim: Drilling and Grooving

Material: Mild Steel

Size: 30 mm x 78 mm

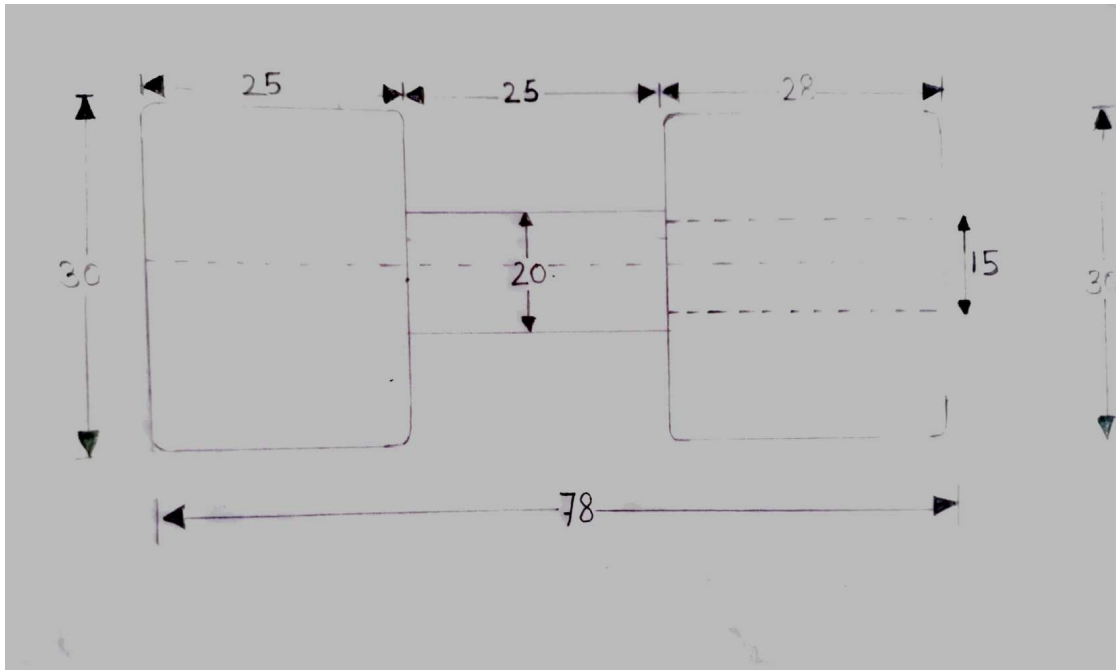
Tolerance: ± 0.06 mm

Tools Required:

- Jaw chuck key
- Tool post key
- Single point cutting tool (High speed steel)
- Grooving Tool
- Drill Chuck Key
- Vernier Caliper
- Straight Shank
- Drill Shaft
- Taper Shank
- File

Job Operation:

1. Job setting
2. Tool setting
3. Facing
4. Plain Turning
5. Drilling
6. Grooving
7. Chamfering
8. Finishing



(All Dimensions are in mm)

Procedure:

1. Firstly, the job is set on jaw chuck.
2. Using the single point cutting tool, the length and radius of the job is reduced to 78mm and 30mm by face turning and plain turning.
3. Now using the vernier caliper the job is marked and divided into 3 parts of 25mm, 25mm and 28mm.
4. The 25mm in the middle is supposed to be grooved out. So, we need to install the grooving tool onto the tool post.
5. Once the grooving tool is installed, we start to groove out the 25mm area, reducing radius of the job area from 30mm to 20mm. We directly cannot measure out the 20mm so we carefully do the grooving part by part and keep on checking the radius using vernier caliper to reach an accurate result.
6. Now the job looks like a dumbbell where one part is 25mm and other part is 28mm.
7. Once grooving is done, we would need to setup the drilling accessories. The straight shank is setup on drill shaft which is tightly set on the tailstock.
8. From right side of the job, which is the 28mm part, we would mark the center point of the job, and use the drill to start drilling a 15 mm thick and 28mm deep hole in it.
9. Once this is done, the pointy edges are now chamfered to rounded edges.
10. The entire job is smoothed out using a sandpaper.
11. Finishing the job.



WORKSHOP

(TURNING EXP-3)

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Experiment 3: To make a job as per drawing

Aim: Threading, Drilling and Grooving

Material: Mild Steel

Size: 80 mm x 40 mm

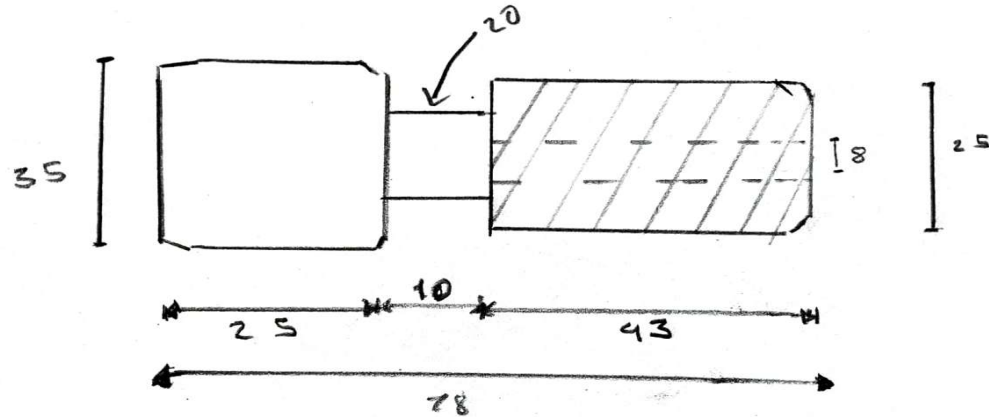
Tolerance: ± 0.04 mm

Tools Required:

- Jaw chuck key
- Tool post key
- Single point cutting tool (High speed steel)
- Grooving Tool
- Drill Chuck Key
- Vernier Caliper
- V Tool
- Straight Shank
- Drill Shaft
- File

Job Operation:

1. Job setting
2. Tool setting
3. Facing
4. Plain Turning
5. Drilling
6. Grooving
7. Threading
8. Chamfering
9. Finishing



(All Dimensions are in mm)

Procedure:

1. Firstly, the job is set on jaw chuck.
2. Using the single point cutting tool, the length and radius of the job is reduced to 78mm and 35mm by face turning and plain turning.
3. Now using the vernier caliper the job is marked and divided into 3 parts of 25mm, 10mm and 43mm.
4. In the right side of the job the 25mm and 18mm diameter is marked using the single point cutting tool.
5. Once marked, from the marked 25mm, plain turning is done for 53mm long, upon reaching we need to groove out the 10mm out of the 53mm.
6. The 10mm in the middle is supposed to be grooved out. So, we need to install the grooving tool onto the tool post.
7. Once the grooving tool is installed, we start to groove out the 10mm area, reducing radius of the job area from 25mm to 20mm. We directly cannot measure out the 20mm so we carefully do the grooving part by part and keep on checking the radius using vernier caliper to reach an accurate result.
8. Once grooving is done, we would need to setup the drilling accessories. The straight shank is setup on drill shaft which is tightly set on the tailstock.
9. From right side of the job, which is the 43mm part, we would mark the center point of the job, and use the drill to start drilling an 8 mm thick and 43mm deep hole in it.

10. Once this is done, the pointy edges are now chamfered to rounded edges.
11. Now we need to do threading on the 43mm part using compound slide tapering method.
12. Using the formula, $\tan \theta = (D-d)/(2L)$, we can find the compound rest angle. And we get 5/8 which is TPI-11 BSW.
13. After setting the machine to gear 5/8, set the machine to automatic which will cause the lead screw to swivel, we should start tapering using single point cutting tool.
14. After each cut, the tool is withdrawn using the cross-slide, brought back to the starting position, and depth is increased gradually. Finally, the threading would be done automatically as per our requirements.
15. Thread pitch is verified using a pitch gauge.
16. The entire job is smoothed out using a sandpaper.
17. Finishing the job.



WORKSHOP

(TURNING EXP-4)

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Experiment 4: To make a job as per drawing

Aim: Internal Threading

Material: Mild Steel

Size: 30 mm x 40 mm

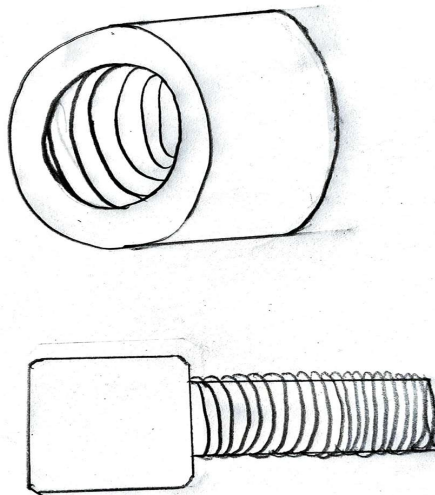
Tolerance: ± 0.04 mm

Tools Required:

- Jaw chuck key
- Tool post key
- Single point cutting tool (High speed steel)
- Drill Chuck Key
- Vernier Caliper
- Cutting Tool
- Boring Tool
- Straight Shank
- Drill Shaft
- Pitch Gauge
- File

Job Operation:

1. Job setting
2. Tool setting
3. Plain Turning
4. Drilling
5. Internal Threading
6. Finishing



(All Dimensions are in mm)

Procedure:

1. Firstly, the job is set on jaw chuck.
2. Use a center drill on the tailstock to drill a short guide hole.
3. A twist drill bit is used to drill a straight hole that goes through the other end of the job.
4. The boring tool is mounted onto the tool post.
5. The compound rest is set at the required angle using the formula $\tan \theta = (D-d)/(2L)$.
6. The lathe is set to the correct gear for 11 TPI.
7. The lead screw is engaged, and the lathe is operated in automatic mode.
8. After each cut, the tool is withdrawn using the cross-slide, brought back to the starting position, and depth is increased gradually.
9. Thread pitch is verified using a pitch gauge.
10. The entire job is smoothed out using a sandpaper.
11. Finishing the job.